Lie Group and Lie Algebra

跟着兹白阿姨学群论

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Part I Group Overview

1. Definition and some properties

First, we give the definition of group:

定义 1..1 – 群 (Group)

A group is a set G of elements $a \in G$ with an operation $\circ$, where $\circ$ is a map:

$$\circ : G \times G \to G, \tag{1.1}$$

$$(a, b) \mapsto a \circ b \equiv ab, \tag{1.2}$$

with the properties:

① **closed**: $\forall a, b \in G$, $ab \in G$.

② **associativity**: $a(bc) = (ab)c$.

③ **identity element**: $\exists e \in G$, $ea = ae = a$, $\forall a \in G$.

④ **inverse element**: $\forall a \in G$, $\exists a^{-1} \in G$, $aa^{-1} = a^{-1}a = e$.

Base on this definition, we have some properties:

(1) Order of a Group G:

$$|G| = \text{number of elements in } G. \tag{1.3}$$

a. $\{e\} : |G| = 1$,

$$ee = e, \quad e^{-1} = e, \quad (ee)e = e(ee) \tag{1.4}$$

b. $G = \mathbb{Z} : |G| = \infty$,

$$a + (-a) = 0, \quad a + 0 = a, \quad (a + b) + c = a + (b + c) \tag{1.5}$$

c. $G = U(1) = \{z \in \mathbb{C} \mid |z| = 1\} : |G| = \infty : z = e^{i\theta}, z^{-1} = z^* = e^{-i\theta}$

$$\Rightarrow e^{i\theta} \cdot e^{-i\theta} = 1, \quad e^{i\theta} \cdot 1 = e^{i\theta}, \quad (e^{i\theta}e^{i\alpha})e^{i\beta} = e^{i\theta}(e^{i\alpha}e^{i\beta}) \tag{1.6}$$

(2) Cayley table: operation rule table

$G = Z_2 = \{e, a\}$

	e	a
e	e	a
a	a	e

$$a^2 = e \text{ or } a^2 = a \Rightarrow a(a - e) = a - e \Rightarrow a = e \Rightarrow \{e\} \Rightarrow a^2 \neq a \tag{1.7}$$

(3) Left/Right translation

$$l_a : G \to G, \quad x \mapsto ax \quad \text{for } a \in G \tag{1.8}$$

$$r_a : G \to G, \quad x \mapsto xa \quad \text{for } a \in G \tag{1.9}$$

l_a and r_a are bijective maps:

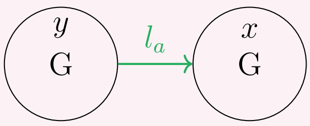
well defined

$x_1 = x_2 \Rightarrow f(x_1) = f(x_2)$.

Surjective (onto)

$f : X \to Y$, $x \mapsto y$ requires image $f(x)$ cover all Y:

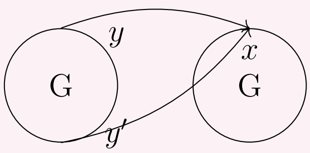
$$l_\forall x \in G : a(y) = x \Rightarrow ay = x \Rightarrow y = a^{-1}x \in G \tag{1.10}$$



Injective (one-to-one):

Every element $x \in X$, $\exists$ a unique $y \in Y$ st. $f(x) = y$:

$$\forall y, y' \in G : l_a(y) = l_a(y') \Rightarrow ay = ay' \Rightarrow y = y' \tag{1.11}$$



(4) Abelian Group

$$\forall a, b \in G \Rightarrow ab = ba \tag{1.12}$$

$E_s = \{e, a, b\}$

	e	a	b
e	e	a	b
a	a	a^2	ab
b	b	ba	b^2

$$\Rightarrow a^2 = e,\ ba = b \quad \text{or} \quad a^2 = b,\ ba = e \tag{1.13}$$

$$a^2 = b, ba = e \Rightarrow ab = aba \Rightarrow a = e \quad \text{(wrong)} \tag{1.14}$$

$$\Rightarrow b^2 = a \cdot a^2 = ea = a. \tag{1.15}$$

$$ab = ba, \Rightarrow Z_3 = \{e, a, a^2\},\ a^3 = e \tag{1.16}$$

(5) Cyclic Group

$$\exists a \in G, \quad G = \{a^k \mid k \in \mathbb{Z}\} \equiv \langle a \rangle \tag{1.17}$$

where $a^{-k} = (a^{-1})^k$, $a^0 := e$, a is a generator of G.

注 1..1. $|a| = |\langle a \rangle|$ is the order of an element $a \in G$.

Any cyclic group is Abelian:

$$a^k \cdot a^l = a^{k+l} = a^{l+k} = a^l \cdot a^k \tag{1.18}$$

due to $k, l \in \mathbb{Z}$ thus $k + l = l + k$.

Generator is not Unique:

例 1..1. $\mathbb{Z}$ is a cyclic group, for every integer n can be

$$n = 1 + 1 + \cdots + 1 = 1^n \quad \text{for } n \geq 1 \tag{1.19}$$

$$n' = (-1) + (-1) + \cdots + (1) = (-1)^n \quad \text{for } n \leq 1 \tag{1.20}$$

$$1^0 = (-1)^0 = 0 \quad \text{for } n = 0. \tag{1.21}$$

but $(1)^n = (1)^{-n} = n \Rightarrow 1$ and -1 both are Generators.

There are only 2 groups in $|G| = 4$: Z_4, K_4

$$Z_4 = \{e, a, a^2, a^3\} \tag{1.22}$$

K_4: symmetry group of rectangle.

$$\Rightarrow K_4 = \{a, b \mid a^2 = b^2 = (ab)^2 = e\} = Z_2 \times Z_2 \tag{1.23}$$

$$= \{e, a, b, ab\} \tag{1.24}$$

(6) Subgroup

For Group G, H, if $H \subseteq G$. then H is a subgroup.

$$H < G. \tag{1.25}$$

with operation $\circ : H \times H \to H$. Trivial subgroup: $\{e\}, G$.

定理 1..1 Subgroup Criterion

$H \neq \emptyset, H \subseteq G$ is a subgroup if and only if

$$ab^{-1} \in H, \quad \forall a, b \in H. \tag{1.26}$$

证明

1. ① necessary: trivial.
2. ② sufficient: $\forall a, b \in H, ab^{-1} \in H$.

   (1) $\forall c \in H, (a \cdot b)c = a \cdot (bc)$, trivial.

   (2) $e \in G \Rightarrow e \in H$.

(3) take $a = e \Rightarrow ab^{-1} = b^{-1} \in H, \forall b \in H$.

(4) $b = (b^{-1})^{-1} \equiv c^{-1} \Rightarrow ab = ac^{-1} \in H$.

例 1..2. $m \in \mathbb{N}, m\mathbb{Z} := \{mk \mid k \in \mathbb{Z}\} \Rightarrow m\mathbb{Z} < \mathbb{Z}$ due to

$$mk + (ml)^{-1} = m(k-l) \in m\mathbb{Z}. \tag{1.27}$$

$m\mathbb{Z}$ is the only subgroup of $\mathbb{Z}$.

(5) Presentation of a group

A group can have multiple presentation ways.

$$Z_n = \langle a \mid a^n = e \rangle = \{a, a^2, \ldots, a^{n-1}, e\}. \tag{1.28}$$

2. Symmetric group and Cayley's theorem

定义 2..1 – Symmetric Group

X is a finite set, then

$$S_X = \{f : X \to X \mid f \text{ is bijective}\} \tag{1.29}$$

is a Symmetric group on X, with the operation being composition.

Symmetric group (Permutation group) is NOT the Symmetry group (transformations leaving object inv.).

- If n is the number of elements of $X \Rightarrow S_X \cong S_n$.
- $|S_n| = n!$, $S_3 \cong D_3$.

定义 2..2

$L_G = \{l_a \mid a \in G\}$ is a group.

- $\forall x \in G, (l_a \circ l_b)(x) = l_a(l_b(x)) = l_a(bx) = abx = l_{ab}(x)$

$$\Rightarrow l_a \circ l_b = l_{ab} \quad \text{(closed)} \tag{1.30}$$

- $l_e x = ex = x \Rightarrow l_e l_a = l_a$
- $(l_a l_{a^{-1}})(x) = a \cdot a^{-1} x = l_e x$
- $l_a(l_b l_c) = (l_a l_b) l_c$

Furthermore, $G \to L_G, a \to l_a$ is bijective,

$$\text{onto} = \text{trivial}$$
$$\text{one-to-one} : l_a = l_b \Rightarrow ax = bx \Rightarrow a = b.$$

Together with this and $l_a l_b = l_{ab}$, we find

$$|G| = |L_G|. \tag{1.31}$$

So G and L_G have the same structure, so

$$G \cong L_G \quad \text{(Isomorphic)} \tag{1.32}$$

This l_a is a permutation operation of G.

$$\Rightarrow l_a \in S_G, \forall a \in G \tag{1.33}$$
$$\Rightarrow L_G \subset S_G \Rightarrow L_G < S_G \tag{1.34}$$

定理 2..1 Cayley's theorem

Any finite group G of order n is isomorphic to a subgroup of S_n:

$$G < S_G. \tag{1.35}$$

$|G| = n, |S_G| = n! \Rightarrow G < S_G$ is a non-trivial subgroup.

3. Cosets and theorem of Lagrange

定义 3..1

$H < G, a \in G$

$$\text{left coset} : aH := \{ax \mid x \in H\} \tag{1.36}$$
$$\text{right coset} : Ha := \{xa \mid x \in H\} \tag{1.37}$$

定理 3..1 Coset Lemma

$a, b \in G, H < G,$

$$aH = bH \iff a^{-1}b \in H \iff b^{-1}a \in H \tag{1.38}$$
$$Ha = Hb \iff ab^{-1} \in H \iff ba^{-1} \in H \tag{1.39}$$

which means the cosets are either $aH = bH$ or $aH \cap bH = \emptyset$.

证明

① let $\forall a, b \in G, H < G$, assume $aH = bH$

$$\begin{aligned}
&\Rightarrow \exists x, y \in H, ax = by \\
&\Rightarrow a = b(yx^{-1}), yx^{-1} \in H \\
&\Rightarrow b^{-1}a = yx^{-1} \in H \Rightarrow a^{-1}b \in H
\end{aligned}$$

② let $a^{-1}b \in H \Rightarrow \exists x \in H, a^{-1}b = x$

$$\begin{aligned}
&\Rightarrow b = ax \Rightarrow \forall y \in H, by = axy = az, z = xy \in H \\
&\Rightarrow bH = aH.
\end{aligned}$$

定义 3..2

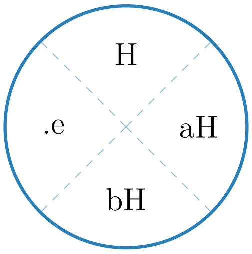
$H < G$,

$$a \sim b := aH = bH \iff a^{-1}b \in H \tag{1.40}$$
$$a \sim b := Ha = Hb \iff ab^{-1} \in H \tag{1.41}$$

It's easy to check it's an equivalence relation:

- $a \sim a : a^{-1}a \in H$
- $a \sim b, a^{-1}b \in H \Rightarrow b^{-1}a \in H \Rightarrow b \sim a$
- $a \sim b, b \sim c, a^{-1}b \in H, b^{-1}c \in H \Rightarrow a^{-1}bb^{-1}c \in H \Rightarrow a^{-1}c \in H \Rightarrow a \sim c$

Thus, G can be separated as a subgroup H and the cosets of H.



定义 3..3

$H < G$

$$G/H := \{aH \mid a \in G\} \quad \text{set of all left cosets} \tag{1.42}$$
$$H\backslash G := \{Ha \mid a \in G\} \quad \text{set of all right cosets} \tag{1.43}$$

定义 3..4 – Index of a Subgroup

$H < G$, The index of H $= |G : H|$ is the number of cosets of H.

Since $l_a : G \to G$ is bijective and since $l_a(H) = aH$,

$$|H| = |aH|, \quad \forall a \in G. \tag{1.44}$$

Thus, we have

定理 3..2 Lagrange's Theorem

G is group, $H < G, |G| < \infty$

$$|G : H| = \frac{|G|}{|H|}. \tag{1.45}$$

If $|G| = P$ is a prime number, then $H = \{e\}$ and $H = G$ are the only two subgroups of G.

例 3..1. $S_3 = D_3 = \{e, a, a^2, b, ab, a^2b\}$, where $a^3 = e, b^2 = e, ba = a^2b, ba^2 = ab$. $H = \{e, a, a^2\} \cong Z_3 \Rightarrow |Z_3| = 3, |D_3| = 6$. $|G : H| = 2$: $\{b, ab, a^2b\}$ is the coset.

$$D_3 = Z_2 \cup \{b, ab, a^2b\}. \tag{1.46}$$

4. Conjugation

定义 4..1

$g \in G, \varphi_g : G \to G, a \mapsto gag^{-1}$ is called the conjugation with g. $a, b \in G$ are conjugate $a \sim b$ if $\exists g \in G : b = gag^{-1}$.

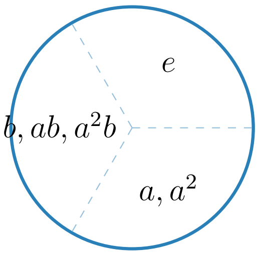
where "$\sim$" is an equivalence relation:

- $a \sim a : eae^{-1} = a$
- $a \sim b \Rightarrow b \sim a : gbg^{-1} = a \Rightarrow b = g^{-1}ag = (g^{-1})a(g^{-1})^{-1}$
- $a \sim b, b \sim c \Rightarrow a \sim c : a = gbg^{-1} = ghch^{-1}g^{-1} = (gh)c(gh)^{-1}$.

定义 4..2 – Conjugation class

$Cl(a) = \{b \in G \mid b \sim a\} = \{gag^{-1} \mid g \in G\}$. For Abelian group: $Cl(a) = \{a\}, \forall g : gag^{-1} = a$.

例 4..1. $D_3 = \{e, a, a^2, b, ab, a^2b\} = \langle a, b \mid a^3 = e, b^2 = e, ba = a^2b\rangle$. $Cl(e) = \{e\}, geg^{-1} = e$. $Cl(a) = \{a, a^2\}$. $Cl(b) = \{b, ab, a^2b\}$.



定义 4..3 – Centralizer

The centralizer of element $a \in G$ is:

$$Cen(a) = \{b \in G \mid ab = ba\}. \tag{1.47}$$

namely all the elements commute with a. For Abelian group $Cen(a) = G$. $Cen(a) < G, \forall a \in G$:

① $b, b' \in Cen(a), bb' \in Cen(a)$.

② $e \in Cen(a)$.

③ $b \in Cen(a), b^{-1} \in Cen(a)$.

定理 4..1

$a \in G$,

$$|G : Cen(a)| = |Cl(a)|. \tag{1.48}$$

$$\text{or } |G| = |Cen(a)||Cl(a)|. \tag{1.49}$$

证明

$|G : Cen(a)| = \frac{|G|}{|Cen(a)|}$, so we need to prove $G/Cen(a) \to Cl(a)$ $A \in Cen(a), g \cdot A \mapsto gag^{-1}$ is a bijective map.

1. ① **well defined**: $g, g' \in G$, suppose $gA = g'B, A, B \in Cen(a)$.

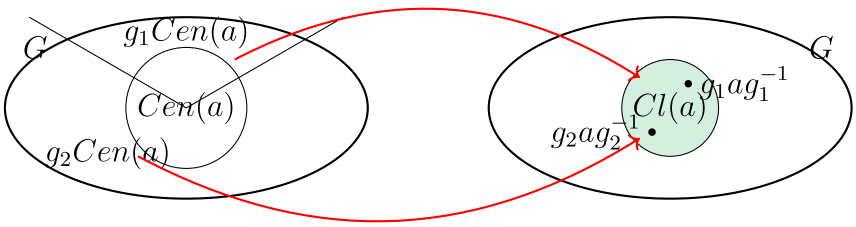
$$\Rightarrow A = g^{-1}g'B, \text{ because } Cen(a) \text{ is a subgroup.}$$
$$\Rightarrow g^{-1}g' \in Cen(a) \Rightarrow (g^{-1}g')a = a(g^{-1}g').$$

Thus, the image of map satisfies: $g'ag'^{-1} = gag^{-1}$.

one-to-one:

$$g', g \in G, \quad g'ag'^{-1} = gag^{-1} \tag{1.50}$$
$$\Rightarrow g^{-1}g'a = a(g^{-1}g') \Rightarrow g^{-1}g' \in Cen(a) \tag{1.51}$$
$$\Rightarrow g^{-1}g'A = B \Rightarrow gB = g'A, \quad A, B \in Cen(a) \tag{1.52}$$

onto: By definition, $\forall b \in Cl(a)$, i.e., $\exists g, b = gag^{-1}$ must be the image of element in $g \cdot Cen(a)$.



If $|G|$ is prime number, G is abelian.
All groups with prime order are cyclic.

定义 4..4 – 中心 (Center of G)

$$Z(G) = \{z \in G \mid az = za, \forall a \in G\} \tag{1.53}$$
$$= \bigcap_{a \in G} Cen(a) \tag{1.54}$$

which represent the set of element $z \in G$ commute with all elements in G.

1. $Z(G) < Cen(a) < G$.
2. $Z(G)$ is Abelian.
3. $Z(G) = G \Rightarrow G$ Abelian.
4. G is centerless: $Z(G) = \{e\}$, e.g. S_3.
5. $z \in Z(G) \Rightarrow Cl(z) = \{z\}$.

Since we separate G as multiple classes $Cl(a_i)$, thus

$$|G| = \sum_i |Cl(a_i)| \tag{1.55}$$

with the theorem

$$|G| = \sum_i [G : Cen(a_i)] \tag{1.56}$$

notice $Cl(z) = \{z\}, \quad z \in Z(G)$

$$\Rightarrow |G| = |Z(G)| + \sum_{a_i \notin Z(G)} [G : Cen(a_i)] \tag{1.57}$$

5. Homomorphisms

定义 5..1 – 同态 (Homomorphism)

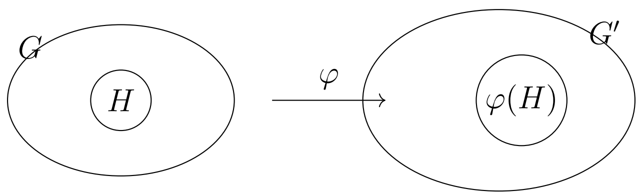
$\varphi : G \to G'$ is a homomorphism if

$$\varphi(ab) = \varphi(a)\varphi(b), \quad \forall a, b \in G. \tag{1.58}$$

注 5..1. The operation in G can be different in G'.

Properties:

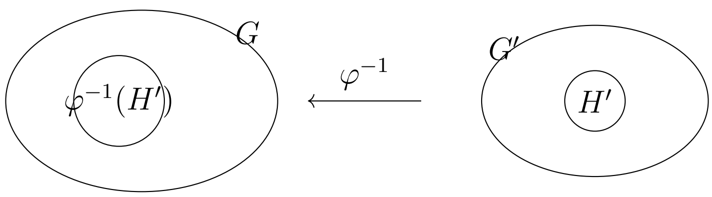
1. $\varphi(e) = e'$
2. $\forall a \in G, \quad \varphi(a^{-1}) = \varphi^{-1}(a)$
3. $H < G \Rightarrow \varphi(H) < \varphi(G) = G'$



where

$$\varphi(H) = \{\varphi(a) \mid a \in H\} \tag{1.59}$$

$H' < G', \quad \varphi^{-1}(H') = \{a \in G \mid \exists a' \in H' = \varphi(a) = a'\}.$



1. $H' < G' \Rightarrow \varphi^{-1}(H') < G$

$$a, b \in \varphi^{-1}(H'), \quad a' = \varphi(a), b' = \varphi(b) \Rightarrow \varphi(a^{-1}b) = a'^{-1}b' \in H' \tag{1.60}$$

$$\Rightarrow a^{-1}b \in \varphi^{-1}(H') \tag{1.61}$$

2. $Im\varphi < G' : Im\varphi = \{\varphi(a) \mid a \in G\} = \varphi(G)$
3. $Ker\varphi < G : Ker\varphi = \{a \in G \mid \varphi(a) = e'\} = \varphi^{-1}(\{e'\})$
4. φ is one-to-one $\iff Ker\varphi = \{e\}$

1. φ is one-to-one, $a \in Ker\varphi \Rightarrow \varphi(a) = e' = \varphi(e)$

$$\Rightarrow a = e \tag{1.62}$$

2. $Ker\varphi = \{e\}$, if $\varphi(a) = \varphi(b) \Rightarrow \varphi(a)\varphi^{-1}(b) = \varphi(ab^{-1}) = e'$

$$\Rightarrow ab^{-1} \in Ker(\varphi) = \{e\} \Rightarrow a = b. \tag{1.63}$$

5. $\varphi_1 : G \to G'$ is homomorphism, $\varphi_2 : G' \to G''$ is homomorphism

$$\Rightarrow \varphi_2 \circ \varphi_1 \text{ is homomorphism} \tag{1.64}$$

$$\varphi_2 \circ \varphi_1(ab) = \varphi_2(\varphi_1(ab)) = \varphi_2(\varphi_1(a)\varphi_1(b)) = \varphi_2\varphi_1(a) \cdot \varphi_2\varphi_1(b) \tag{1.65}$$

定义 5..2

$\varphi : G \to G'$ is homomorphism.

- If $G = G' \Rightarrow$ **endomorphism**.
- φ bijective $\Rightarrow$ **isomorphism**.
- $G = G'$ and φ bijective $\Rightarrow$ **automorphism**.

$$Aut(G) := \{\varphi : G \to G \mid \varphi \text{ is bijective homomorphism}\} \tag{1.66}$$

$$Aut(G) < S_G \tag{1.67}$$

定理 5..1

Conjugation is automorphism:

$$\varphi_g : G \to G : \varphi_g(a) = gag^{-1} \text{ is automorphism.} \tag{1.68}$$

For $H < G, \varphi_g(H) \cong H$.

定理 5..2

G is cyclic.

$$\text{if } |G| = \infty, \quad G \cong \mathbb{Z} \tag{1.69}$$

$$|G| = m, \quad G \cong Z_m \tag{1.70}$$

6. Normal subgroups and factor groups

定义 6..1

G is group, $N < G$, N is a **normal subgroup**

$$N \triangleleft G \tag{1.71}$$

if $aN = Na, \forall a \in G$, namely left and right cosets are the same. So we find

$$N = aNa^{-1} \Rightarrow \text{invariant under conjugation.} \tag{1.72}$$

定义 6..2

$N \triangleleft G$,

$$G/N = \{aN \mid a \in G\} \text{ with } aN \circ bN = (ab)N \tag{1.73}$$

is the factor group of G modulo N.

定理 6..1

G and G' are groups. $\varphi : G \to G'$ is homomorphism.

- $Ker\varphi \triangleleft G$
- $N \triangleleft G \Rightarrow \varphi(N) \triangleleft \varphi(G) < G'$

证明

① $Ker\varphi = \{a \in G \mid \varphi(a) = e'\}$

$$\forall g \in G, \quad a \in Ker\varphi \tag{1.74}$$

$$\varphi(gag^{-1}) = \varphi(g)\varphi(a)\varphi(g^{-1}) = \varphi(g)e'\varphi^{-1}(g) = e' \tag{1.75}$$

$$\Rightarrow gag^{-1} \in Ker\varphi \Rightarrow gKer\varphi g^{-1} \subset Ker\varphi \tag{1.76}$$

$$\Rightarrow gKer\varphi \subset Ker\varphi g \quad \text{and} \quad Ker\varphi g^{-1} \subset g^{-1}Ker\varphi \tag{1.77}$$

$$\Rightarrow gKer\varphi = Ker\varphi g. \tag{1.78}$$

② $\varphi(G) < G', \quad \varphi(N) < \varphi(G)$

$$\forall g' \in \varphi(G), \quad n' \in \varphi(N) \tag{1.79}$$

$$g'n' = \varphi(g)\varphi(n) = \varphi(gn) = \varphi(mg) \text{ (where } m, n \in N) \tag{1.80}$$

where $n, m \in N$, thus normal subgroup:

$$gN = Ng \Rightarrow gn = mg \tag{1.81}$$

$$\Rightarrow g'n' = m'g' \Rightarrow g'\varphi(n) \in g'\varphi(N) \Rightarrow g'\varphi(N) \subset \varphi(N)g' \tag{1.82}$$

$$n'g'^{-1} = g'^{-1}m' \Rightarrow \varphi(N)g'^{-1} \subset g'^{-1}\varphi(N) \Rightarrow g'\varphi(N) = \varphi(N)g' \tag{1.83}$$

All subgroups in Abelian group are normal.

定义 6..3

G is **simple**, if $\{e\}$ and G are the only normal subgroups.

7. Product of groups

First isomorphism theorem:

$$\varphi : G \to G' \text{ is homomorphism,} \tag{1.84}$$

define $\bar{\varphi} = G/Ker\varphi \to Im\varphi < G'$

$$a \cdot Ker\varphi \to \varphi(a) \tag{1.85}$$

implies $\bar{\varphi}$ is isomorphism, and $G/Ker\varphi \cong Im\varphi$.

证明

① **homomorphism**: $\forall a, b \in G$,

$$\bar{\varphi}(aKer\varphi)\bar{\varphi}(bKer\varphi) = \varphi(a)\varphi(b) = \varphi(ab) = \bar{\varphi}(abKer\varphi) \tag{1.86}$$

$$= \bar{\varphi}(aKer\varphi \cdot bKer\varphi) \tag{1.87}$$

② **one-to-one**: $\forall a, b \in G, \quad aKer\varphi = bKer\varphi$

$$\bar{\varphi}(aKer\varphi) = \varphi(a), \quad \bar{\varphi}(bKer\varphi) = \varphi(b) \tag{1.88}$$

$$\bar{\varphi}(a^{-1}bKer\varphi) = \varphi(a^{-1}b) = \varphi^{-1}(a)\varphi(b) = e' \tag{1.89}$$

due to $a^{-1}bKer\varphi = Ker\varphi \Rightarrow a^{-1}b \in Ker\varphi$

$$\Rightarrow \varphi(a) = \varphi(b) \Rightarrow \bar{\varphi}(aKer\varphi) = \bar{\varphi}(bKer\varphi) \tag{1.90}$$

③ **onto**: $\forall a, b \in G, \quad \varphi(a) = \varphi(b) \Rightarrow \varphi(a^{-1}b) = e'$

$$\Rightarrow a^{-1}b \in Ker\varphi \Rightarrow a^{-1}bKer\varphi = Ker\varphi \Rightarrow \bar{\varphi}(aKer\varphi) = \bar{\varphi}(bKer\varphi) \tag{1.91}$$

定义 7..1

G_1, G_2 are groups. The Cartesian product

$$G_1 \times G_2 = \{(a_1, a_2) \mid a_1 \in G_1, a_2 \in G_2\} \tag{1.92}$$

with the operation

$$\circ : (G_1 \times G_2) \times (G_1 \times G_2) \to G_1 \times G_2 \tag{1.93}$$

$$((a_1, a_2), (b_1, b_2)) = (a_1 b_1, a_2 b_2) \tag{1.94}$$

is the direct product of G_1 with G_2.

- G_1, G_2 are Abelian $\iff$ $G_1 \times G_2$ is Abelian.
- $G_1 \times G_2 \neq G_2 \times G_1$, but $G_1 \times G_2 \cong G_2 \times G_1$, due to

$$(a_1, a_2) \mapsto (a_2, a_1) \text{ is isomorphism.} \tag{1.95}$$

例 7..1. ① $Z_2 \times Z_2 \cong K_4$

$$Z_2 = \{e, a\}, \quad K_4 = \{e, a, b, ab\} = \{a, b \mid a^2 = b^2 = e\} \tag{1.96}$$

② $Z_3 \times Z_2 \cong Z_2 \times Z_3 \cong Z_6$

$$Z_2 = \{e_1, a\}, \quad Z_3 = \{e_2, b, b^2\} \tag{1.97}$$

$$Z_2 \times Z_3 = \{(a, b) \mid a \in Z_2, b \in Z_3\} \tag{1.98}$$

$$= \{(e_1, e_2), (e_1, b), (e_1, b^2), (a, e_2), (a, b), (a, b^2)\} \tag{1.99}$$

$$= \{e, b, b^2, a, ab, ab^2\} = \langle a, b \mid a^2 = e, b^3 = e, ab = ba\rangle \tag{1.100}$$

$$\Rightarrow ab \cdot ab = a^2b^2 = b^2, \quad (ab)^3 = e \tag{1.101}$$

$$(ab)^4 = b, \quad (ab)^5 = ab^2, \quad (ab)^6 = e \tag{1.102}$$

$$\Rightarrow Z_2 \times Z_3 \cong Z_6 \tag{1.103}$$

定理 7..1

$\{e_1\}, \{e_2\}$ is the trivial subgroups of G_1, G_2. now:

$$G_1' = G_1 \times \{e_2\} = \{(g_1, e_2) \mid g_1 \in G_1\} \tag{1.104}$$

$$G_2' = \{e_1\} \times G_2 = \{(e_1, g_2) \mid g_2 \in G_2\} \tag{1.105}$$

obviously $G_{1,2}' \cong G_{1,2}, \quad G_{1,2}' < G_1 \times G_2$ we claim: $G_{1,2}' \lhd G_1 \times G_2$

$$G_1 \times G_2/G_{1,2}' \cong G_{2,1} \tag{1.106}$$

证明

we first define the projection onto G_2'

$$k_2 : G_1 \times G_2 \to G_1 \times G_2 \tag{1.107}$$

$$(g_1, g_2) \mapsto (e_1, g_2) \tag{1.108}$$

it's a homomorphism:

$$(g_1, g_2) \cdot (h_1, h_2) = (g_1h_1, g_2h_2) \tag{1.109}$$

$$(e_1, g_2) \cdot (e_1, h_2) = (e_1, g_2h_2) \tag{1.110}$$

thus $Kerk_2 = \{(g_1, e_2) \mid g_1 \in G_1\} = G_1' \lhd G_1 \times G_2$ (**Theorem 4**) Base on the first isomorphism theorem.

$$\exists\bar{\varphi} : G_1 \times G_2/G_1' \to G_1 \times G_2 \tag{1.111}$$

$$(g_1', g_2')(g_1, e_2) \mapsto (h_1, h_2) \tag{1.112}$$

namely $(g_1'g_1, g_2'e_2) \in G_1 \times G_2$. thus, $G_1 \times G_2/G_1' \cong Imk_2 \cong G_2$

The necessary conditions for factorization of a group G.

1. $G_1' \cap G_2' = \{(e_1, e_2)\}$
2. $G_1'G_2' = G$.
3. $a_1'a_2' = a_2'a_1'$, for $a_1' \in G_1', a_2' \in G_2'$
4. $G_1' \lhd G, \quad G_2' \lhd G$

①②③ or ①②④ are also sufficient.

定理 7..2

$G_1 < G, G_2 < G$, with

- $G_1 \cap G_2 = \{e\}$
- $G_1 G_2 = \{a_1 a_2 \mid a_1 \in G_1, a_2 \in G_2\} = G$
- $G_1 \lhd G, \quad G_2 \lhd G$

then: $G_1 \times G_2 \cong G$ namely: $G_1 \times G_2 \to G$

$$(a, b) \mapsto ab \text{ is isomorphism.} \tag{1.113}$$

further more: $a_1 a_2 = a_2 a_1, \forall a_1 \in G_1, a_2 \in G_2$.

例 7..2. $K_4 = \langle a, b \mid a^2 = b^2 = (ab)^2 = e \rangle$

$$Z_2 \cap Z_2 = e, \quad Z_2 \lhd K_4 \text{ due to } K_4 \text{ abelian} \tag{1.114}$$

$$Z_2 \cdot Z_2' = \{ab \mid a \in Z_2, b \in Z_2'\} \tag{1.115}$$

$$= \{e, a, b, ab\} = K_4 \tag{1.116}$$